ANDREW STEVENS

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Website Portfolio: www.andrewstevens.engineering

EDUCATION

Virginia Tech– Blacksburg, VA **Expected May 2029**
B.S. Electrical Engineering - Junior standing by credits **GPA: 4.0**
Selected Courses: Linear Algebra, Digital Systems, Computational Eng, Electromagnetic Physics, Differential Equations

PROFESSIONAL EXPERIENCE

Internship - Computer Science **May 2026 - August 2026**
Hitachi Energy – Raleigh, NC
- Scoped and planned an autonomous Python report generator, defining requirements and milestones
- Built the application to parse transformer service test files and generate formatted reports, reducing documentation time from 4–6 hours to ~5 minutes (~98% reduction)

Current Owner And Founder **March 2021 - Present**
Andy's 3D Workshop – Ashburn, VA
- Established the company and developed business plans for optimized sales growth
- Created the company's online store on Etsy with over 1,600 sales and a 5-star rating
- Designed and prototyped all products, including 1 patented with US Patent & Trademark Office
- Sold products to businesses and partnered with 2 local stores in Northern Virginia

Substitute Teacher **August 2025 - Present**
Loudoun County Public Schools – Ashburn, VA
- Maintain continuity of learning by implementing lesson plans and ensuring a productive classroom environment
- Adapt teaching strategies to meet the diverse needs of students throughout grades K-12

ADDITIONAL EXPERIENCE

Connected & Autonomous Vehicles (CAV) Team Member - (Onboarding) **January 2026 - Present**
HEVT – Virginia Tech
- Designed Cooperative Adaptive Cruise Control in simulink to increase energy efficiency of Cruise Control by 60.3%
- Improved existing Python code of ROS2 Cruise Control to account for the changed Cooperative Adaptive Cruise Control
- Training in high-voltage safety systems, powertrain architecture, vehicle sensing, control integration, and cross functional engineering processes for an electric vehicle that is part of the EcoCar Competition
- Designed and implemented intersection crossing logic in Simulink and control blocks to govern vehicle decision-making

ECE Subteam **September 2025 - Present**
VT CRO Workcell – Virginia Tech
- Designed an automated calibration system in Python with computer vision to better 3D print precision and efficiency
- Developed an automated material handling system to have filament runout detection and multi-material 3D printing
- Designing a printer UI and control system to interface with the printers, providing status feedback and parameter control

Testing & Electronics **September 2025 - March 2026**
Baja SAE – Virginia Tech
- Developed a custom microcontroller GPS tracker to track position and velocity, to improve race performance analysis
- Made a Python interface to visualize sensor data at any point during a race, improving diagnostics and evaluation
- Developing telemetry for real time monitoring of vehicle health in competitions, improving reliability and safety
- Learned Ki-CAD to design custom professional PCBs which improves the system reliability for the Baja electronics suite

Engineering Lead **August 2023 - June 2025**
Vex Robotics – Academies Of Loudoun
- Led the engineering design and documentation for a VEX Robotics team qualified for the State Championship
- Applied prototyping and project management skills to build competitive robots and guide strategy to the State level

SKILLS & INTERESTS

- Technical Skills: 3D CAD Modeling, Certified in Arduino Circuitry, C#, C++, Python, Microsoft Suite, MATLAB
- Soft Skills: Easily Adaptable, Self-Motivation, Problem Solving, Project Management, Teamwork, Time Management
- Personal Projects: Developed a BLE-controlled ESP32 RC car controlled by an iOS app, a Python-based system using YOLOv8 for custom control via hand tracking, and “The Cube”, an Arduino-powered interactive puzzle box integrating sensors, displays, and logic-based progression